



SATHARAKA NILMANTHA

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GitHub

LinkedIn

Portfolio

▼ ABOUT ME

Junior Associate Software Developer with a strong Computer Engineering foundation, dedicated to building innovative and high-performance systems. I thrive on translating real-world problems into technical solutions, ensuring scalability and reliability at every stage of the development lifecycle. Actively engaged in expanding my technical stack to drive product innovation and operational efficiency.

▼ EDUCATION

BSc. Engineering (Hons.) in Computer Engineering

Apr 2022 – Aug 2026

University of Jaffna, Sri Lanka

Relevant Coursework: Software Engineering, Object-Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Computer Architecture, Embedded Systems Design, Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Advanced Networking.

G.C.E. Advanced Level – Physical Science Stream

2016–2020

Dhammissara National Collage, Nattandiya

Results: A – Combined Mathematics, B – Chemistry, C – Physics

▼ WORK EXPERIENCES

Software Developer - Junior Associate

May 2026- Present

Cyber Mas Solutions (Pvt) Ltd., Colombo, Sri Lanka

VETAHR Payroll Management System (*Tech Stack: Node.js, Express.js, React.js, Tailwind CSS, MySQL, Docker, Nginx, JWT, Hikvision API*)

- **Maintain** and enhance **payroll system** ensuring stability, performance, and compliance. Optimize queries and improve **fingerprint matching** via **Hikvision API** for accurate attendance.
- **Develop new features** and **RESTful APIs** with **Node.js + Express.js + JWT**. Build **responsive UI** with **React.js + Tailwind CSS** aligning with business needs.
- **Manage deployment** with **Docker + Nginx**. Collaborate with stakeholders as key **Junior Associate**, continuing development from internship.

Software Developer Intern

November 2025 – May 2026

Cyber Mas Solutions (Pvt) Ltd., Colombo, Sri Lanka

Smart Dispatch System (*Tech Stack: Java Spring Boot, React.js, Tailwind CSS, MySQL, Docker, Nginx, Grafana, JMeter, JWT*)

- **Built** microservices dispatch platform with **barcode generation/scanning** for instant package retrieval. Developed **React.js + Tailwind CSS** frontend consuming **REST APIs** via **Spring Cloud API Gateway**.
- **Architected Spring Boot** microservices with **Spring Cloud** for **service discovery** and **API gateway**. Implemented **JWT-based authentication + HTTPS security** across all services, with **MySQL** following **database-per-service isolation**.
- **Deployed** via **Docker + Nginx** on **Ubuntu**, integrated **Grafana** for monitoring and tested with **Apache JMeter** to validate **scalability** and **performance** under peak load.

VETAHR Payroll Management System (*Tech Stack: Node.js, Express.js, React.js, Tailwind CSS, MySQL, Docker, Nginx, JWT, Hikvision API*)

- **Developed** automated **payroll system** handling attendance, leave requests, and payroll processing with **ETF/EPF contributions** and **bank salary transfers**. Integrated **fingerprint matching** via **Hikvision API** for real-time biometric attendance.
- **Built secure RESTful APIs** using **Node.js + Express.js** with **JWT-based authentication** and authorization. Developed **responsive UI** with **React.js + Tailwind CSS**, collaborating in an **Agile team** environment.
- **Deployed** stack using **Docker + Nginx** on cloud infrastructure with managed **cloud database**, ensuring seamless scalability and reliable performance.

▼ PROJECTS

HealthHub – Smart Hospital Management System

Jan 2025 - May 2025

Tech Stack: Java Spring Boot, React.js, MySQL, Bootstrap, Postman, Jira

- Contributed to full-stack development of **role-based hospital management system** using **Spring Boot + React.js**. Implemented **admin employee management** and **receptionist access** to doctor prescriptions.
- Designed **responsive UI** for patients, doctors, admins, and receptionists. Implemented **real-time appointment booking** and **doctor prescription features** for streamlined workflows.
- Conducted **QA testing** with manual + exploratory methods, **validated APIs** using **Postman**, and collaborated in **Agile sprints** tracked via **Jira**.

Smart Pet Feeder – IoT-Based Automated Feeding System

Jan 2025- May2025

Tech Stack: ESP32, C, React Native, Firebase

- Developed **IoT pet feeder** using **ESP32 + C** with **ultrasonic sensors** for **real-time food level monitoring** and **instant alerts** via **Firebase** notifications.
- Built **React Native mobile app** for **remote control**, **feeding schedule management**, **manual override**, and **dynamic portion control** with **offline support**.
- Utilized **Firebase** for **scheduling logic**, **data storage**, and **device-user synchronization**, preventing food waste and overfeeding.

BookMyDoctor – Doctor Appointment Booking Management System

Nov 2024 - Jun 2025

Tech Stack: Java Spring Boot, React.js, MySQL, Twilio, Bootstrap, bcrypt, File Handling, Chatbot Integration

- Developed **web-based appointment system** with **secure authentication** using **bcrypt**, **file handling**, and **responsive UI** with **Bootstrap**.
- Integrated **Twilio** for **SMS notifications** including confirmations, cancellations, reminders, and **next-day alerts** for patients.
- Built **admin features** for appointment management, doctor registration/activation, and patient notifications. Integrated **React-based chatbot** for doctor details (specialization, availability, contact).

Machine Learning-Based Intrusion Detection System for Public Wi-Fi Networks

Jan 2025 - Aug 2026

Tech Stack: Machine Learning, Python, Scikit-Learn, XGBoost, Pandas, NumPy, Scapy, pfSense, Mininet, Network Security

- Developed a **real-time IDS** to detect **ARP Spoofing & MITM attacks** on public Wi-Fi using supervised ML, engineering **15 behavioral features** via **Borda Count consensus** (XGBoost Gain, Mutual Information, ANOVA, Permutation Importance).
- Deployed **SmartIDS plugin** on **pfSense** with **dual-layer defense** (heuristic + XGBoost), **model quantization** for **<1ms inference**, ensuring **100% offline**, **privacy-first** operation with **zero internet slowdown**.
- Simulated attacks via **Mininet**, benchmarked against **Arp watch & Snort**, optimized **XGBoost pipeline** with **5-fold cross-validation** and early stopping, achieving **high accuracy with minimal false positives** for **edge deployment**.

▼ SKILLS

- Programming Languages** : Java, Python, C++, JavaScript, HTML, CSS
- Frameworks & Libraries** : React.js, Spring Boot, Node.js, Bootstrap, Tailwind CSS
- Databases** : MongoDB, MySQL, Firebase
- Tools & Technologies** : Git, GitHub, Postman, Figma, Twilio, Docker, Nginx, Grafana, JMeter
- Network & Security** : Wireshark, WebRTC, pfSense
- Soft Skills** : Team Collaboration, Problem Solving, Self-Learning, Leadership

▼ VOLUNTEERING EXPERIENCES

IESL Student Member

Jan 2023 – Present

As a Student Member of the Institution of Engineers, Sri Lanka (IESL), I actively engage with the engineering community, gaining exposure to professional practices, technical knowledge, and industry standards through the institution's events and resources

Member No: S-31553

▼ REFERENCES

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